# Day 5 Assignment

**Description:**

This project implements a frontend blogging platform where users can:

- View blog posts retrieved from a remote server
- View todo items retrieved from a remote server
- Publish their own text posts
- Manage tasks using a floating todo interface

The project uses the **Fetch API** to retrieve data from remote endpoints and implements proper error handling to ensure application stability. The code is structured using the **Revealing Module Pattern** to improve maintainability and scalability.

**Objective:**

The primary objectives of this assignment were:

- To gain hands-on experience with the Fetch API
- To work with JSON data from public APIs
- To dynamically manipulate the DOM
- To implement robust error handling mechanisms
- To understand and apply a JavaScript design pattern

**Setup and API Understanding:**

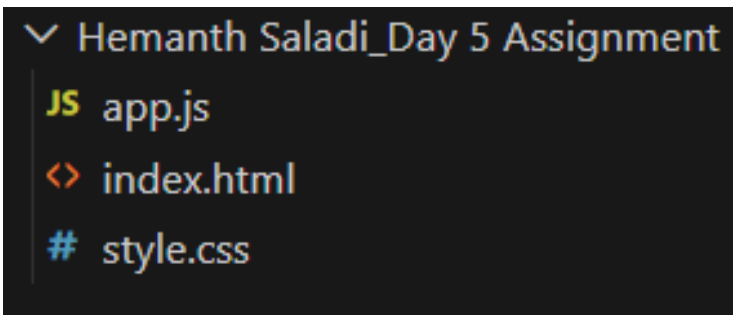
The application uses the public testing API:

- Base URL: https://jsonplaceholder.typicode.com

Endpoints used:

- /posts → Retrieves blog posts
- /todos → Retrieves todo item

**Folder Structure:**

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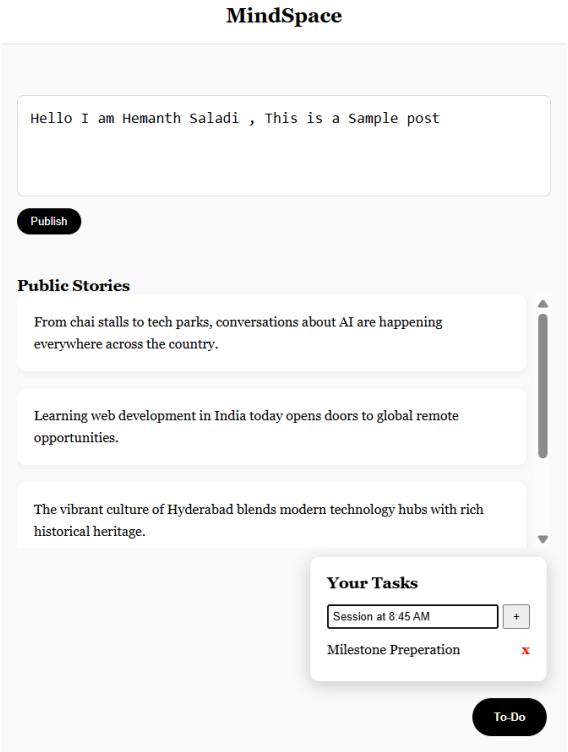
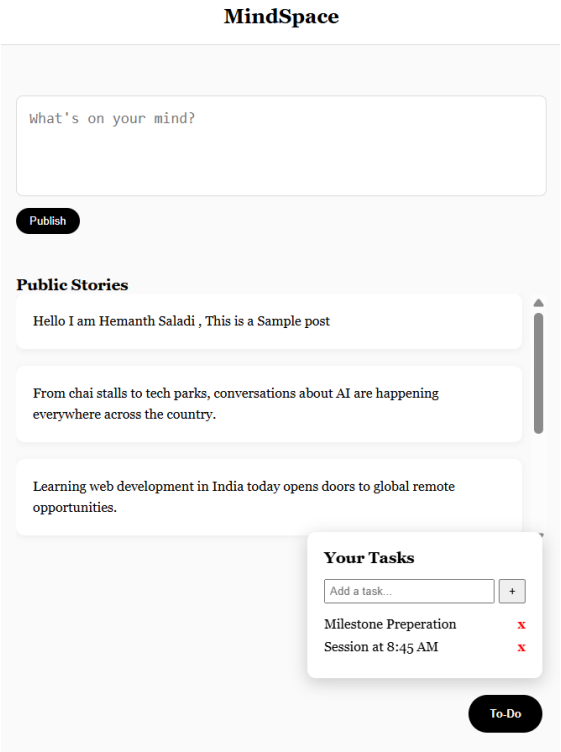
## Task 1: Fetch and Display Posts:

Blog posts were retrieved from the /posts endpoint using the Fetch API with async/await. The server response was validated and converted into JSON format. The posts were dynamically displayed on the webpage using DOM manipulation.

## Task 2: Fetch and Display Todos:

Todo items were fetched from the /todos endpoint. After validating the response and converting it to JSON, the data was rendered dynamically. Completed tasks were displayed differently from pending tasks for better clarity.




## Task 3: Error Handling:

Error handling was implemented using **try-catch blocks** and **response.ok** validation. In case of network failure or invalid responses, user-friendly error messages were shown to prevent application crashes.

## Task 4: JavaScript Design Pattern:

The Revealing Module Pattern was used to organize the code. This helped encapsulate logic, avoid global variables, and improve maintainability and scalability.

## Task 5: Integration and Testing:

All features including data fetching, rendering, and error handling were integrated into a single cohesive application. The application was tested to ensure correct data retrieval and smooth user interaction.

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**Challenges Faced:**

1. Handling asynchronous operations properly.
2. Validating server response correctly.
3. Structuring code without global variables.
4. Managing UI updates dynamically.

Code Snippets:

*Index.html :*

Hemanth Saladi_Day 5 Assignment > index.html > ...

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>MindSpace - Share Your Thoughts</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

<header class="navbar">
    <h2>MindSpace</h2>
</header>

<main class="container">

    <!-- Post Input Section -->
    <section class="post-input">
        <textarea id="userPost" placeholder="What's on your mind?"></textarea>
        <button id="postBtn">Publish</button>
    </section>

    <!-- Scrollable Public Posts Section -->
    <section class="posts-section">
        <h3>Public Stories</h3>
        <div id="postsContainer" class="posts-container"></div>
    </section>

</main>

<!-- Floating To-Do Button -->
<button class="todo-btn" id="todoToggle">To-Do</button>

<!-- Todo Popup -->
<div class="todo-popup" id="todoPopup">
    <h3>Your Tasks</h3>

    <div class="todo-input">
        <input type="text" id="taskInput" placeholder="Add a task..." />
        <button id="addTask">+</button>
    </div>

    <ul id="taskList"></ul>
</div>

<script src="app.js"></script>
</body>
</html>
```

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*App.js:*

Hemanth Saladi_Day 5 Assignment > JS app.js > ...

```javascript
// BLOG + TODO MODULE
const App = (function () {

    const postBtn = document.getElementById("postBtn");
    const userPost = document.getElementById("userPost");
    const postsContainer = document.getElementById("postsContainer");

    const todoToggle = document.getElementById("todoToggle");
    const todoPopup = document.getElementById("todoPopup");
    const addTaskBtn = document.getElementById("addTask");
    const taskInput = document.getElementById("taskInput");
    const taskList = document.getElementById("taskList");

    /* Preloaded Public Posts (India Context) */
    const samplePosts = [
        "India's digital revolution is reshaping how young innovators build startups from small towns.",
        "The vibrant culture of Hyderabad blends modern technology hubs with rich historical heritage.",
        "Learning web development in India today opens doors to global remote opportunities.",
        "From chai stalls to tech parks, conversations about AI are happening everywhere across the country."
    ];

    function loadPublicPosts() {
        samplePosts.forEach(text => {
            createPost(text);
        });
    }

    function createPost(content) {
        const post = document.createElement("div");
        post.classList.add("post");
        post.textContent = content;
        postsContainer.prepend(post);
    }

    function publishPost() {
        const content = userPost.value.trim();
        if (content) {
            createPost(content);
            userPost.value = "";
        }
    }

    function toggleTodo() {
        todoPopup.style.display =
            todoPopup.style.display === "block" ? "none" : "block";
    }

    function addTask() {
        const task = taskInput.value.trim();
        if (!task) return;

        const li = document.createElement("li");
        li.innerHTML = `
            ${task}
            <span>x</span>
        `;

        li.querySelector("span").onclick = () => li.remove();

        taskList.appendChild(li);
        taskInput.value = "";
    }

    function init() {
        loadPublicPosts();
        postBtn.addEventListener("click", publishPost);
        todoToggle.addEventListener("click", toggleTodo);
        addTaskBtn.addEventListener("click", addTask);
    }

    return { init };

})();

document.addEventListener("DOMContentLoaded", App.init);
```

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